

QuantumPhysEval: Measuring Quantum Hallucinations in LLMs

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Abstract

This paper presents QuantumPhysEval, a benchmark developed to measure whether large language models produce physically inconsistent yet superficially plausible answers on elementary quantum-mechanical tasks. QuantumPhysEval evaluates model behavior in four task families central to introductory quantum reasoning: circuit evolution, operator composition, measurement prediction, and entanglement classification. Because the benchmark is built from analytically generated two-qubit problems, it supports exact self-generated ground truth and fully automated scoring with bounded normalized physics-error metrics. Through a case study involving seven GPT-4 and GPT-5 family models across three prompted reasoning-depth settings, QuantumPhysEval identifies clear reliability gaps and offers practical insight into where those models fail. In the pooled analysis of 5,376 evaluations, benchmark error decreases strongly with a model-size proxy, with fitted exponent $\alpha = 2.257$ and bootstrap 95% interval $[1.933, 2.587]$, while prompted reasoning depth shows only a weak effect with $\beta = -0.137$ and interval $[-0.523, 0.258]$. A separate 64-task repair ablation over stronger models shows that one-shot solving reaches 0.422 solved fraction, blind self-retry reaches 0.609, and verifier-guided retry reaches 0.688, with nearly all gains saturating within the first 30 seconds. The most important practical observation is that the largest residual risks are concentrated in circuit evolution and measurement prediction, indicating that exact quantum dynamics remain difficult even for stronger recent models.

1 Introduction

Large Language Models (LLMs) have shown strong performance on many reasoning and coding tasks, but there are still relatively few benchmarks designed to measure whether their outputs remain faithful to exact scientific structure. In quantum information settings, this matters because correct answers must satisfy sharply defined physical constraints such as normalization, unitarity, linear state evolution, and the Born rule. An answer can therefore look polished and plausible while still being physically wrong.

In this work, we identify one major reliability risk for language models in this domain: *quantum hallucination*, which we define as a model output that is syntactically plausible but inconsistent with exact quantum-mechanical ground truth or with basic physical constraints. To measure this risk, we develop QuantumPhysEval, an evaluation suite intended to integrate directly into model testing workflows in the same way that safety and robustness benchmarks are used in other domains.

1.1 QuantumPhysEval Approach

QuantumPhysEval’s overall approach is straightforward. First, we construct benchmark prompts from analytically generated two-qubit tasks so that exact targets are known by construction. Second, we score model outputs using bounded physics-grounded error metrics rather than subjective grading. Third, we apply the resulting benchmark to multiple model families and prompted depth settings in order to understand which kinds of quantum tasks remain difficult and whether stronger models reduce those errors.

This benchmark is designed to be useful in practice. Because test generation, target generation, and evaluation are automated, the suite can be rerun easily as models improve. Because it uses exact two-qubit simulation, every failure corresponds to a real physical inconsistency rather than a matter of interpretation.

1.2 Main Contributions

Overall, QuantumPhysEval makes the following contributions:

- **Exactness:** every test case has an exact self-generated quantum target derived analytically inside the harness.

QuantumPhysEval pipeline

The benchmark is self-labeled: every prompt has exact quantum ground truth, so model error is measured directly rather than judged heuristically.

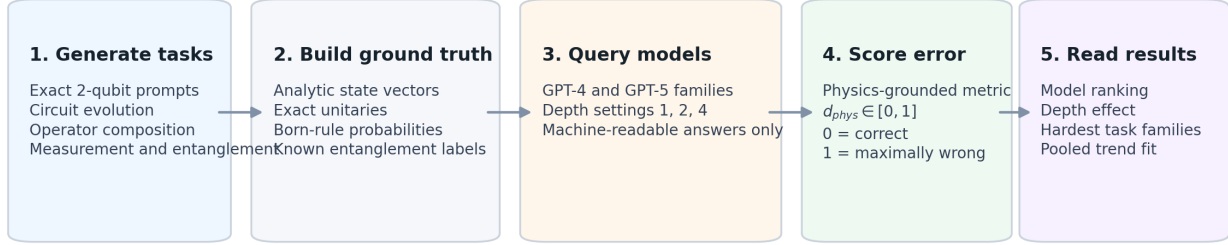


Figure 1: QuantumPhysEval benchmark overview. The pipeline is fully automated: exact two-qubit tasks are generated analytically, exact targets are computed by construction, model outputs are scored with a bounded physics-grounded error metric, and pooled trends are analyzed across model families and depth settings.

- **Breadth:** the benchmark covers four distinct quantum reasoning families: circuit evolution, operator composition, measurement prediction, and entanglement classification.
- **Automation:** test generation, model querying, parsing, scoring, and scaling-law fitting are fully automated.
- **Actionability:** in a pooled GPT-4/GPT-5 case study, the benchmark identifies concrete reliability gaps and isolates the task families that drive them.

1.3 The structure of this paper

The rest of the paper is organized as follows. First, we describe QuantumPhysEval’s benchmark construction and scoring methodology. Next, we describe the case study protocol used to evaluate pooled GPT-4 and GPT-5 family models. We then present top-level results, category-level findings, and ablations. Finally, we discuss limitations and conclude.

2 Benchmark Construction

QuantumPhysEval contains four benchmark families:

- **Circuit evolution:** generate the final two-qubit state after a sampled gate sequence.
- **Operator composition:** generate the exact one-qubit unitary corresponding to a sampled gate sequence.
- **Measurement prediction:** generate the exact post-circuit measurement probabilities.
- **Entanglement classification:** decide whether a provided two-qubit state is entangled.

2.1 Constructing test cases

All benchmark instances are generated analytically inside the evaluation harness. The gate library consists of single-qubit gates $\{I, H, X, Y, Z, S, T\}$ and two-qubit gates $\{\text{CNOT}(0, 1), \text{CNOT}(1, 0), \text{CZ}\}$. For circuit evolution and measurement prediction, the benchmark samples initial states and gate sequences and then computes exact outcomes. For operator composition, it samples one-qubit gate sequences and asks the model to return the corresponding unitary. For entanglement classification, it evaluates analytically specified product and Bell-like states.

Each run contains 64 prompts per benchmark family, yielding 256 unique prompts per run. Because the benchmark is generated from exact quantum primitives rather than mined from free-form text, the task distribution is controlled and reproducible.

2.2 Generating exact ground truth

Ground truth is produced directly from quantum mechanics rather than external annotation. States and operators are computed by explicit matrix multiplication. Measurement targets are derived from exact Born-rule probabilities. Entanglement labels are attached by construction. This allows the benchmark to avoid noisy labels and to support exact automated evaluation.

3 Methods

Each model response is parsed into a structured object and scored with a bounded physics-grounded error metric $d_{phys} \in [0, 1]$, where 0 indicates a correct response and 1 indicates a maximally bad or unparseable response under the benchmark metric.

3.1 Scoring state, operator, probability, and boolean outputs

For circuit-evolution tasks, the predicted state is normalized, aligned up to global phase, and compared to the exact target using a combination of normalization penalty and one minus fidelity. For measurement tasks, negative values are clipped to zero, the output is renormalized, and the score combines total-variation distance with a normalization penalty. For operator-composition tasks, the score averages a unitarity penalty with a target-mismatch penalty based on normalized Frobenius norms. For entanglement classification, the score is 0 for an exact match and 1 otherwise.

These metrics are intentionally simple. The goal is not to model every nuance of how a human instructor would grade a quantum mechanics homework answer; the goal is to determine whether the model returned an object that is physically consistent with the exact target.

3.2 Case study protocol

To study benchmark behavior in practice, we apply QuantumPhysEval to two completed runs. The first evaluates `gpt-4o-mini`, `gpt-4.1-mini`, and `gpt-4.1`. The second evaluates `gpt-5.1`, `gpt-5.2`, `gpt-5.4-mini`, and `gpt-5.4`. Each model is tested at prompted reasoning-depth settings $D \in \{1, 2, 4\}$ with temperature fixed at 0.0. The system prompt enforces machine-readable answers only, and the user prompt requests the specified number of internal reasoning steps while returning only the final answer.

Across both runs, the pooled evaluation contains 5,376 scored responses. To summarize aggregate scaling behavior, we fit

$$\mathbb{E}[d_{phys}] = kS^{-\alpha}D^{\beta},$$

using a model-size proxy S and log-linear least squares, with 2,000 bootstrap resamples for uncertainty estimation.

3.3 Time-budget repair ablation

We also run a separate repair ablation designed to test whether the hardest failures are recoverable with additional interaction time. This ablation uses a strong-model subset consisting of `gpt-4.1`, `gpt-5.2`, `gpt-5.4-mini`, and `gpt-5.4`; depth 4 only; temperature 0.2; and four prompts per benchmark family, for 64 tasks total. Each prompt receives a five-minute wall-clock budget and is evaluated under three modes: *one-shot* (no retry), *self-retry* (the model takes another pass without being told whether the prior answer was wrong), and *verifier feedback* (the model is told only that the prior answer was incorrect against exact ground truth).

For this ablation we report solved fraction by deadline (pass@30s, pass@60s, pass@120s, pass@300s), mean and median time-to-correctness among solved prompts, and category- and model-level solved fractions. The goal is to distinguish failures that are fixable with iterative inference from failures that persist even under repeated repair attempts.

4 Results

4.1 Top-level findings

The pooled mean normalized error is 0.440, with median 0.354. The 90th and 95th percentiles are both 1.0, indicating a heavy tail of severe failures. Exact-correct outputs occur on 31.7% of samples, while 29.3% of

Table 1: Mean normalized physics error by model in the pooled case study. Lower is better.

Model	Mean d_{phys}
gpt-5.4	0.329
gpt-5.2	0.368
gpt-5.4-mini	0.394
gpt-4.1	0.442
gpt-4.1-mini	0.478
gpt-5.1	0.501
gpt-4o-mini	0.570

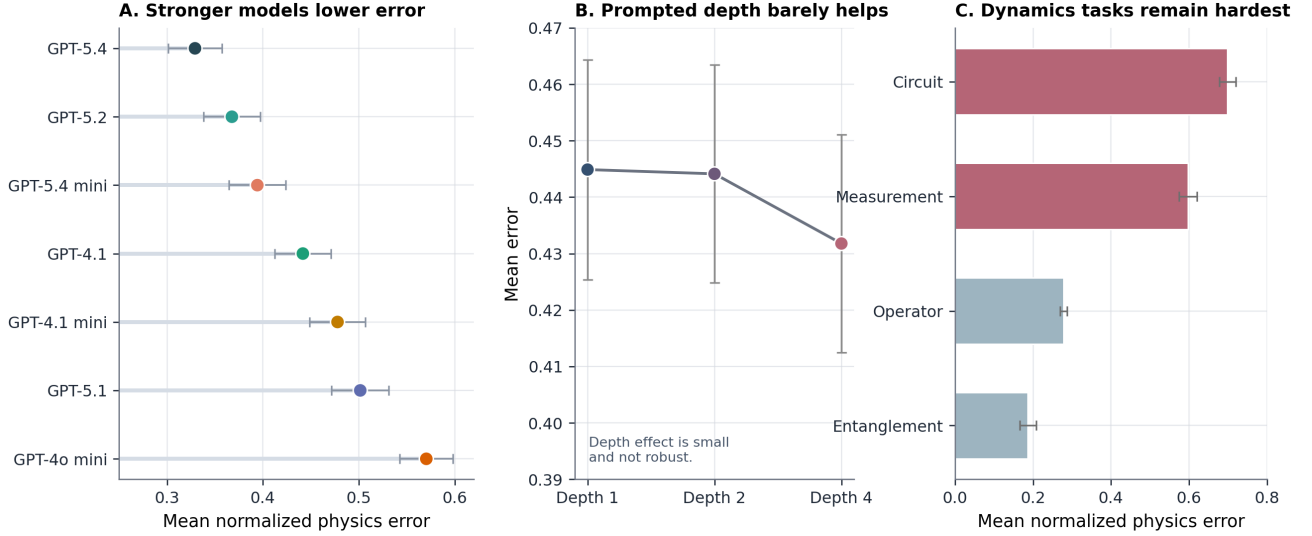


Figure 2: Direct takeaways from the pooled GPT-4/GPT-5 case study. Left: stronger models reduce physics-grounded error. Middle: more prompted reasoning depth changes the result only slightly. Right: the hardest residual failures remain circuit evolution and measurement prediction.

samples are effectively full-error failures. QuantumPhysEval is therefore measuring a substantial reliability gap rather than just small calibration drift.

As shown in Table 1, the strongest model in the pooled case study is **gpt-5.4** and the weakest is **gpt-4o-mini**. The ordering is not perfectly monotone with the manually assigned size proxy, but the overall trend is clear: more capable models reduce physics-grounded benchmark error.

The following theses emerged from the pooled benchmark results:

1. All studied models exhibit quantum hallucinations at non-trivial rates.
2. More capable models reduce those errors, but do not eliminate them.
3. Prompted reasoning depth has only a weak effect relative to model capability.
4. The largest failures are concentrated in exact quantum dynamics tasks.

4.2 Scaling behavior

The strongest quantitative pattern in the pooled analysis is the size effect. Fitting the pooled data yields

$$\alpha = 2.257 \quad \text{with bootstrap 95\% CI [1.933, 2.587].}$$

Under the current proxy assignment, this supports a strong decrease in benchmark error with model capability or scale.

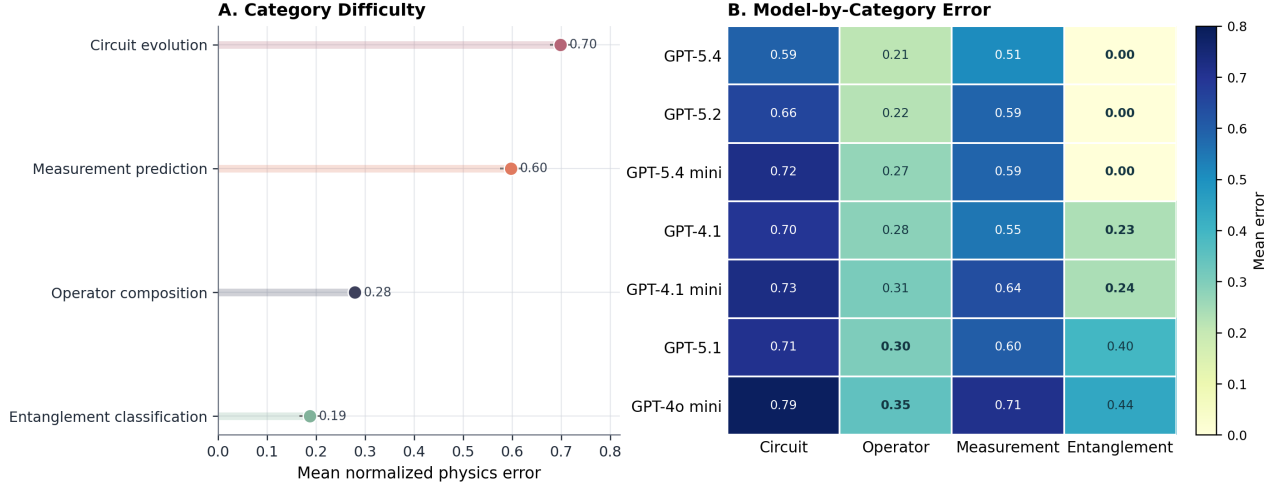


Figure 3: Benchmark-family breakdown in the pooled case study. Left: overall category difficulty. Right: model-by-category heatmap showing where each model fails most strongly.

The prompted reasoning-depth effect is much weaker:

$$\beta = -0.137 \quad \text{with bootstrap 95\% CI } [-0.523, 0.258].$$

Average error at depth 4 is 0.432, compared with 0.445 at depth 1 and 0.444 at depth 2. We therefore do not find robust evidence that increasing the prompted depth condition alone materially improves benchmark performance.

4.3 Benchmark-family findings

Benchmark-family differences are pronounced. In the pooled run, category means are:

- **Circuit evolution:** 0.698
- **Measurement prediction:** 0.597
- **Operator composition:** 0.278
- **Entanglement classification:** 0.187

The dominant residual risks therefore arise in tasks that require exact quantum dynamics. Models are substantially better at entanglement classification and operator composition than they are at propagating amplitudes through gates or generating exact measurement distributions.

This pattern also appears at the model-family level. The easiest cells in the pooled run are entanglement classification for `gpt-5.2`, `gpt-5.4`, and `gpt-5.4-mini`, each of which achieves zero mean error on that benchmark slice. The hardest cells remain circuit evolution for weaker models, with `gpt-4o-mini` reaching 0.786 mean error and `gpt-4.1-mini` reaching 0.728.

4.4 Ablations

Additional ablations sharpen the main picture. First, the depth manipulation is weak not only in aggregate but also within individual models: most model-specific depth curves are nearly flat, and where differences do exist they are small relative to cross-model gaps. Second, GPT-5-family models improve over GPT-4-family models in every benchmark category, with the largest gain in entanglement classification and the smallest gain in circuit evolution. Third, stronger models are not simply converting catastrophic failures into milder failures; they also increase exact-correct rates while reducing full-error failures.

4.4.1 Time-budget repair and feedback

The repair ablation adds a more operational view of the benchmark. As shown in Table 2, repeated attempts help substantially, but verifier-guided retry helps more than blind self-retry. Solved fraction rises

Table 2: Feedback ablation on a strong-model subset under a 5-minute per-prompt budget. The subset contains 64 tasks total, depth 4 only.

Mode	Solve rate	Mean solve time (s)	Median solve time (s)
One-shot	0.422	0.835	0.732
Self-retry	0.609	2.650	1.288
Verifier feedback	0.688	2.350	1.025

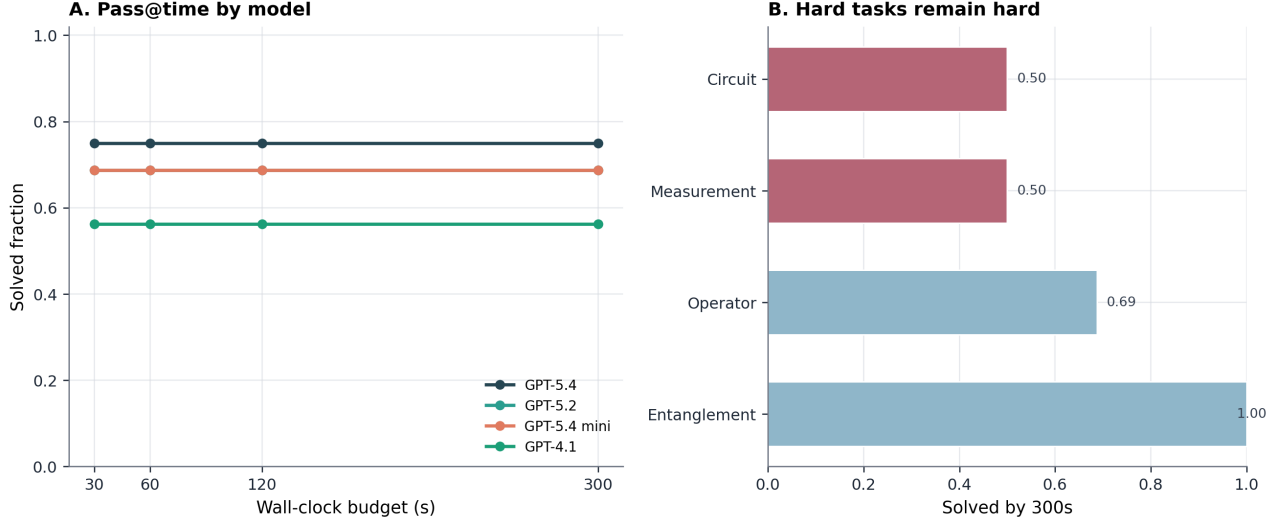


Figure 4: Verifier-feedback time-budget run on the strong-model subset. Left: solved fraction by wall-clock budget for each model. Right: solved fraction by category under the full 300-second budget. In practice the curves plateau early: pass@30s, pass@60s, pass@120s, and pass@300s are identical at 0.688 overall.

from 0.422 in the one-shot condition to 0.609 under self-retry and 0.688 under verifier feedback. Verifier feedback is also slightly more efficient on solved prompts: its mean solve time is 2.350s versus 2.650s for self-retry, and its median solve time is 1.025s versus 1.288s.

The gain is not uniform across task families. Under the same 64-task subset, circuit evolution improves from 0.250 solved fraction in the one-shot setting to 0.500 with verifier feedback; measurement prediction improves from 0.250 to 0.562; and operator composition improves from 0.250 to 0.688. Entanglement classification is already nearly saturated in the one-shot condition (0.938) and reaches 1.000 under verifier feedback. These category-level improvements show that some hard failures are repairable, but they also preserve the same ordering seen in the main benchmark: the dynamics-heavy tasks remain the limiting cases.

Figure 4 makes the key point visible: nearly all recoverable gains happen very early. In the verifier-feedback run, pass@30s, pass@60s, pass@120s, and pass@300s are all equal to 0.688. This means the remaining failures are not being rescued by simply allowing more wall-clock time. The evidence therefore suggests a mixed picture: iterative repair helps, explicit feedback helps more, but extra time alone does not solve the core quantum-dynamics failure modes.

5 Discussion

The case study supports four practical conclusions.

First, quantum hallucination is measurable in a fully automated setting. Many outputs are syntactically clean and machine-readable, yet remain physically wrong under exact evaluation. This makes the benchmark useful as a reliability diagnostic rather than merely as a formatting test.

Second, model capability matters more than prompted depth in the present setup. The strongest result is the pooled capability trend; the depth manipulation is small relative to that effect and does not separate

from zero under bootstrap uncertainty.

Third, some failures are recoverable under iterative interaction, but the mechanism matters. Blind self-retry improves solve rate over a one-shot baseline, while verifier-guided retry improves it further. At the same time, the time-budget curves flatten quickly, which indicates that the remaining hard cases are not simply waiting for more deliberation time.

Fourth, the benchmark identifies a concrete residual weakness: exact quantum state and probability evolution. This is more informative than a generic claim that models are unreliable on quantum tasks, because it isolates the specific computational burden that remains difficult even for stronger models.

6 Limitations

QuantumPhysEval is intentionally narrow. It uses a controlled two-qubit setting to enable exact scoring, but that also limits ecological breadth. The model-size variable is a manually assigned proxy rather than a verified parameter count, and the pooled fit mixes families and release generations. The benchmark also uses a single prompt style per task family and a single temperature setting. The repair ablation is smaller than the pooled study and uses a binary verifier signal that would not always be available in practical deployments. The present evidence therefore supports a strong benchmark-level capability trend, but it should not be overinterpreted as a universal scaling law.

7 Conclusion

In this paper, we introduced QuantumPhysEval, a benchmark for measuring quantum hallucinations as physics-grounded error on elementary quantum tasks. In a pooled case study over GPT-4 and GPT-5 family models, benchmark error decreases strongly with model capability, while prompted reasoning depth shows no robust independent effect. A separate repair ablation shows that repeated attempts can recover some failures, especially when verifier feedback is available, but those gains saturate quickly and do not erase the main weakness. The largest remaining risks are concentrated in circuit evolution and measurement prediction. These results suggest that language models can produce plausible quantum answers, but exact physical reliability remains limited and should be measured explicitly rather than assumed.

Appendix: Live Statistical Snapshot

Appendix Snapshot

This appendix summarizes the pooled GPT-4/GPT-5 benchmark over 5376 scored responses. The full machine-generated statistics remain available separately in `combined_quantum_results_live.tex`. The pooled fit yields $\alpha = 2.257$ with 95% interval $[1.933, 2.587]$ and $\beta = -0.137$ with interval $[-0.523, 0.258]$.

Top-Line Summary

Metric	Value
Usable evaluations	5376
Mean d_{phys}	0.440
Median d_{phys}	0.354
Exact-correct rate	0.317
$d_{phys} \geq 0.5$ rate	0.435
Full-error rate	0.293
Best model	GPT-5.4
Weakest model	GPT-4o mini

Model Ranking

Model	Mean	95% CI	Exact	Full
GPT-5.4	0.329	[0.301, 0.357]	0.430	0.195
GPT-5.2	0.368	[0.338, 0.397]	0.380	0.249
GPT-5.4 mini	0.394	[0.364, 0.424]	0.372	0.267
GPT-4.1	0.442	[0.412, 0.471]	0.311	0.293
GPT-4.1 mini	0.478	[0.449, 0.507]	0.285	0.311
GPT-5.1	0.501	[0.472, 0.531]	0.260	0.355
GPT-4o mini	0.570	[0.543, 0.598]	0.177	0.378

Depth and Category Breakdown

By depth				By category			
Depth	Mean	Exact	Full	Category	Mean	Exact	Full
1	0.445	0.312	0.301	Circuit	0.698	0.165	0.502
2	0.444	0.315	0.291	Measurement	0.597	0.225	0.475
4	0.432	0.323	0.285	Operator	0.278	0.063	0.006
				Entanglement	0.187	0.813	0.187

Model-by-Category Means

Model	Circuit	Measurement	Operator	Entanglement
GPT-5.4	0.592	0.509	0.215	0.000
GPT-5.2	0.659	0.587	0.224	0.000
GPT-5.4 mini	0.721	0.586	0.270	0.000
GPT-4.1	0.697	0.552	0.283	0.234
GPT-4.1 mini	0.728	0.635	0.308	0.240
GPT-5.1	0.707	0.602	0.302	0.396
GPT-4o mini	0.786	0.711	0.347	0.438

Time-Budget Feedback Ablation

We also ran a separate repair ablation on a strong-model subset (`gpt-4.1`, `gpt-5.2`, `gpt-5.4 mini`, `gpt-5.4`) using depth 4 only, temperature 0.2, four prompts per family, and a five-minute per-prompt budget. Three repair modes were compared: one-shot, self-retry without explicit error signals, and verifier-guided retry.

Mode	Solve rate	Mean solve time	Median solve time
One-shot	0.422	0.835 s	0.732 s
Self-retry	0.609	2.650 s	1.288 s
Verifier feedback	0.688	2.350 s	1.025 s

In the verifier-feedback run, `pass@30s`, `pass@60s`, `pass@120s`, and `pass@300s` were all equal to 0.688, indicating that recoverable gains saturated very early.

Category	One-shot	Self-retry	Verifier feedback
Circuit	0.250	0.375	0.500
Measurement	0.250	0.500	0.562
Operator	0.250	0.625	0.688
Entanglement	0.938	0.938	1.000

Model	One-shot	Self-retry	Verifier feedback
GPT-4.1	0.312	0.438	0.562
GPT-5.2	0.500	0.688	0.750
GPT-5.4 mini	0.438	0.500	0.562
GPT-5.4	0.438	0.812	0.875